Introduction to prismatic cohomology

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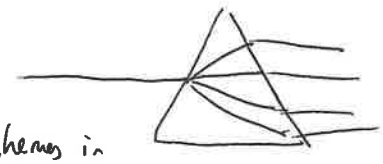
<u>Lecture 1</u>. Fix prime p



schemes in mixed char

prisms

crystalline
de Rham
Hodge-Tate
étale

cohom. theories

$X/\mathbb{Z}_p$ smooth p-adic formal scheme $\rightsquigarrow R\Gamma_{crys}(X_s/\mathbb{Z}_p)$, $X_s = X \times_{\mathrm{Spf}\,\mathbb{Z}_p} \mathrm{Spec}\,\mathbb{F}_p$

$\rightsquigarrow R\Gamma_{dR}(X/\mathbb{Z}_p) = R\Gamma(X, \mathcal{O}_X \xrightarrow{d} \Omega^1_{X/\mathbb{Z}_p} \xrightarrow{d} \cdots)$

$\rightsquigarrow R\Gamma_{HT}(X)$ = twisted form of $R\Gamma(X, \mathcal{O}_X \xrightarrow{0} \Omega^1_{X/\mathbb{Z}_p} \xrightarrow{0} \cdots)$

$\rightsquigarrow R\Gamma_{\acute{e}t}(X_\eta, \mathbb{Z}_p)$. X_η rig. gen. fibre

§ Prismatic cohom. in char 0

k field of char 0, $A = k[[t]]$, R k-alg.

<u>Def'n</u>. (prismatic site in char 0): $(R/A)^{op}_{\Delta}$ with objects

$$\left(R \xrightarrow{\iota} S/t \leftarrow S\right)$$ with S = t-complete, t-torsion free A-alg

& $R \xrightarrow{\iota} S/t$ map of k-alg

- morphisms $\left(R \xrightarrow{\iota} S/t \leftarrow S\right) \to \left(R \xrightarrow{\iota'} S'/t \leftarrow S'\right)$ given by $\alpha: S \to S'$ map of A-alg, s.t. $\alpha \circ \iota = \iota'$.

covers given by isom's.

Set $\mathcal{O}_{\Delta}(R \to S/t \leftarrow S) := S$, $\overline{\mathcal{O}}_{\Delta}(R \to S/t \leftarrow S) = S/t$

$$R\Gamma\left((R/A)_{\Delta}, \mathcal{O}_{\Delta}\right) = R\varprojlim_{(R \to S/t \leftarrow S)} S$$

<u>Th.</u> (Waßmuth) R smooth over k.

1) $H^i\left((R/A)_{\Delta}, \overline{\mathcal{O}}_{\Delta}\right) \underset{can}{\simeq} \Omega^i_{R/k}$, $i \geq 0$

2) $\tilde{R}$ t-complete, t-torsion free lift of R to A, then

$$R\Gamma\left((R/A)_{\Delta}, \mathcal{O}_{\Delta}\right) = \left[\tilde{R} \xrightarrow{t\text{-}d} \hat{\Omega}^1_{\tilde{R}/A} \xrightarrow{t\text{-}d} \hat{\Omega}^2_{\tilde{R}/A} \to \cdots\right]$$

Sketch of proof:

Step 1. comparison map in 1).

Have SES $0 \to \overline{\mathcal{O}}_{\Delta} \xrightarrow{t} \mathcal{O}_{\Delta/t^2} \to \overline{\mathcal{O}}_{\Delta} \to 0$

$\rightsquigarrow$ get $H^0\left((R/A)_{\Delta}, \overline{\mathcal{O}}_{\Delta}\right) \xrightarrow{\beta} H^1\left((R/A)_{\Delta}, \overline{\mathcal{O}}_{\Delta}\right)$

By definition $\uparrow c_{abs}$

R

Now, $H^*\left((R/A)_{\Delta}, \overline{\mathcal{O}}_{\Delta}\right)$ is a commutative graded alg. &

$\beta \circ c_{abs}$ is a k-linear derivation.

$\Rightarrow$ get unique extension $d_{HT} : \Omega^*_{R/k} \to H^*\left((R/A)_{\Delta}, \overline{\mathcal{O}}_{\Delta}\right)$

$\rightsquigarrow$ reduce 1) to 2)

Step 2. A formula for $R\Gamma((R/A)_{\Delta}, \mathcal{O}_{\Delta})$

<u>Lemma</u> $(R/A)_{\Delta}$ has fibre products (via $\hat{\otimes}$) and non-empty product.

To get products

<u>Def'n</u> T t-complete, t-torsion free A-alg $J \subset T$ any ideal, $t \in J$.

The prismatic envelope $D_J(T)$ of T in J is $T[\frac{j}{t} : j \in J]^{\wedge}_t$.

<u>Note</u> S t-complete, t-tors. free A-alg, then

$$\mathrm{Hom}((T, J), (S, (t))) \simeq \mathrm{Hom}((D_J(T), (t)), (S, (t)))$$

Now, let $(R \xrightarrow{\iota} S/t \leftarrow S)$, $(R \xrightarrow{\iota'} S'/t \leftarrow S')$

Set $T = S \hat{\otimes}_A S'$, $J = \ker(T \to T/t = S/t \otimes_R S'/t \to S/t \otimes_R S'/t)$

Then $(R \xrightarrow{\iota \otimes \mathrm{Id}} T/t \to D_J(T)/t \leftarrow D_J(T))$ represents the product.

Thus, $R\Gamma((R/A)_{\Delta}, \mathcal{O}_{\Delta})$ can be calculated as follows.

Pick P t-complete smooth A-alg with surj. $P \twoheadrightarrow R$, having kernel J.

Set $D = D_J(P)$, then $(R \to D/t \leftarrow D) \in (R/A)_{\Delta}$ covers the final object of $\widetilde{(R/A)_{\Delta}}$. $\rightsquigarrow R\Gamma((R/A)_{\Delta}, \mathcal{O}_{\Delta}) = (D(0) \to D(1) \to D(2) \to \cdots)$

$D(i) = (i+1)$-fold product of D in $(R/A)_{\Delta}$

Note: $D(n) =$ prismatic envelope of $P \hat{\otimes}_A (n+1)$ and $J_n = \ker(P \hat{\otimes} n+1 \twoheadrightarrow R)$

& $J_n = (t, x_1, \cdots, x_r)$ gen. by reg. sequence. (locally)

$\Rightarrow D(n) = P \hat{\otimes}(n+1) [\frac{x_1}{t}, \cdots, \frac{x_r}{t}]^{\wedge}_t$ & $D(n)/t \simeq R[\frac{x_1}{t}, \cdots \frac{x_r}{t}]$

Step 3. t- de Rham complexes

Indeed,

$$\begin{tikzcd} P \arrow[r, "t\cdot d"] \arrow[d] & \hat{\Omega}^1_{P/A} \hat{\otimes}_P D \\ D \arrow[ur, dashed, "\exists!\, \nabla^t"] & \end{tikzcd}$$

$$\nabla^t\left(\frac{j}{t}\right) = dj. \quad \text{for } j \in J$$

Step 4. Finish the argument

Set $P(n) = P \hat{\otimes}_A (n+1)$

$$\begin{tikzcd} \vdots & & \\ \hat{\Omega}^1_{P(0)/A} \hat{\otimes}_P D \arrow[r] & & \\ D(0) \arrow[u, "\nabla^t"] \arrow[r] & D(1) \arrow[u] \arrow[r] & \cdots \end{tikzcd}$$

Set $K^{a,b} = \hat{\Omega}^b_{P(a)/A} \hat{\otimes}_{P(a)} D(a)$

Rather formal: $K^{\cdot,b}$ acyclic for $b>0$.

$$\Rightarrow \quad \mathrm{Tot}(K^{\cdot,\cdot}) \simeq \left(D(0) \to D(1) \to \cdots\right)$$

Claim: each of the cosimplicial str. maps $D(0) \to D(n)$ induces a q isom.

$K^{0,\cdot} \to K^{n,\cdot}$.

$$\rightsquigarrow \quad \mathrm{Tot}(K^{\cdot,\cdot}) = R\lim_{\Delta} K^{n,\cdot} = K^{0,\cdot} \underset{\substack{D = P = \tilde{R} \\ \tilde{R}/t = R}}{\simeq} \left(\tilde{R} \xrightarrow{t\cdot d} \hat{\Omega}^1_{\tilde{R}/A} \xrightarrow{t\cdot d} \cdots\right)$$

Lemma S t- complete, t- torsion free A- alg

$P = S[x_1, \dots, x_r]^{\wedge}_t$, $J = (t, x_1, \dots, x_r)$, $D = D_J(P)$.

$$\Rightarrow 0 \to S \to D \xrightarrow{\nabla^t} \hat{\Omega}^1_{P/S} \hat{\otimes}_P D \to \cdots \quad \text{is exact}$$

Proof: reduce mod t.

$$D/t \simeq S/t\left[\frac{x_1}{t}, \dots, \frac{x_r}{t}\right], \qquad \nabla^t\left(\frac{x_i}{t}\right) = dx_i$$

$\rightsquigarrow$ apply usual Poincaré Lemma for poly. rings in char. 0.

§ prismatic cohomology in char. p

Fix prime p. All rings are p-complete.

Def. A δ-ring is a ring A with a map $\delta = \delta_A : A \to A$ (of sets), s.t.

i) $\delta(0) = 0$, $\delta(1) = 0$

ii) $\delta(xy) = x^p \delta(y) + y^p \delta(x) + p\,\delta(x)\delta(y)$.

iii) $\delta(x+y) = \delta(x) + \delta(y) + \dfrac{x^p + y^p - (x+y)^p}{p}$

Given δ-ring A, set $\varphi = \varphi_A : A \to A$, $x \mapsto x^p + p\,\delta(x)$

Lemma 1) A δ-ring $\Rightarrow$ φ_A is a lift of Frobenius

2) A p-torsion free, $\varphi : A \to A$ lift of Frobenius

$$\delta(x) = \frac{\varphi(x) - x^p}{p} \quad \text{is a } \delta\text{-str. on } A$$

Fix a perfect field k, R k-alg.

Def. (prismatic site in char p) $(R/W(k))_{\Delta}$ has objects

$(R \xrightarrow{\iota} A/p \leftarrow A)$ with A p-complete, p-tors. free, δ-ring over $W(k)$.

morph = morphism of δ-rings over $W(k)$ comp. with ι

topology $A \to B$ cover if $A/p \to B/p$ is faithfully flat.

Set $\mathcal{O}_\Delta(R \to A/p \leftarrow A)$, $\bar{\mathcal{O}}_\Delta(R \to A/p \leftarrow A)$

$$\begin{array}{ccc} \mathcal{O}_\Delta(R \to A/p \leftarrow A) & & \bar{\mathcal{O}}_\Delta(R \to A/p \leftarrow A) \\ \| & & \| \\ A & & A/p \end{array}$$

<u>Th.</u> (Bhatt-Scholze, Ogus) R smooth over k.

1) $\varphi^*_{W(k)} R\Gamma((R/W(k))_\Delta, \mathcal{O}_\Delta) \simeq R\Gamma_{crys}(R/W(k))$

2) $H^*((R/W(k))_\Delta, \bar{\mathcal{O}}_\Delta) \simeq \Omega^*_{R/k}$ canonically

3) If $\tilde{R}$ p-completely smooth δ-lift of R to $W(k)$, then

$$R\Gamma((R/W(k))_\Delta, \mathcal{O}_\Delta) \simeq \left[\tilde{R} \xrightarrow{p\cdot d} \hat{\Omega}^1_{\tilde{R}/W(k)} \xrightarrow{p\cdot d} \cdots\right] \quad \varphi\text{-equiv.}$$

<u>Rmk.</u> $\Omega^*_{R/k} \longrightarrow H^*((R/W(k))_\Delta, \bar{\mathcal{O}}_\Delta)$ can be constr. as before

in 3)

$$\begin{tikzcd} R\Gamma((R/W(k))_\Delta, \mathcal{O}_\Delta) \arrow[d] & \simeq & \left[\tilde{R} \arrow[r, "d"] \arrow[d, "\varphi_{\tilde{R}}"] & \frac{1}{p}\hat{\Omega}^1_{\tilde{R}/W(k)} \arrow[r, "d"] \arrow[d, "\varphi_{\tilde{R}}"] & \frac{1}{p^2}\hat{\Omega}^2_{\tilde{R}/W(k)} \arrow[r] \arrow[d, "\varphi_{\tilde{R}}"] & \cdots\right] \\ \varphi_{W(k),*} R\Gamma((R/W(k))_\Delta, \mathcal{O}_\Delta) & \simeq & \varphi_{W(k),*}\left[\tilde{R} \arrow[r, "d"] & \hat{\Omega}_{\tilde{R}/W(k)} \arrow[r, "d"] & \cdots\right] & \end{tikzcd}$$

Modulo p reduces to Cartier isom.

$$\Omega^i_{R^{(1)}/k} \xrightarrow{\sim} H^i(\Omega^*_{R/k}) \quad \text{for } R^{(1)} = R \otimes_{R,\varphi} k$$

Some ingredients in the proof of 1)

Lemma. $|-|: \{\delta\text{-rings}\} \longrightarrow \{\text{rings}\}$ has a left adjoint F & a right adjoint G.

In fact, $G(R) = W(R)$ = p-typical Witt vectors of R

& $\mathbb{Z}\{x\} := F(\mathbb{Z}[x]) = \mathbb{Z}[x, \delta(x), \delta^2(x), \dots]$

Prop. Given p-compl., p-tors free δ-$W(k)$-alg P & $J = (p, x_1, \dots, x_r)$ gen by a reg sequence. then $D := P\{\frac{x_1}{p}, \dots, \frac{x_r}{p}\}_p^\wedge$ is the prismatic envelope of P in J.

$$\mathrm{Hom}\big((P, J), (A, (p))\big) = \mathrm{Hom}\big((D, (p)), (A, (p))\big)$$

δ-pairs p-compl., p-tors. free

Key Lemma: A p-torsion free δ-$\mathbb{Z}_p$-alg, $x \in A$, s.t. $\frac{\varphi(x)}{p} \in A$, then x has all divided powers.

In fact, if $(p, x_1, \dots, x_r)$ is regular, then

$$A\left\{\frac{\varphi(x_1)}{p}, \dots, \frac{\varphi(x_r)}{p}\right\} = \text{PD-envelope of } (x_1, \dots, x_r) \text{ in } A.$$

Sketch. $\frac{\varphi(x)}{p} = \frac{x^p}{p} + \varphi(x) \rightsquigarrow \gamma_p(x) = \frac{x^p}{p!} \in A$

Claim $\gamma_{p^2}(x) = \underset{\mathbb{Z}_p}{\text{unit}} \cdot \frac{x^{p^2}}{p^{p+1}} \in A$

$$A \ni \delta\left(\frac{x^p}{p}\right) = \frac{1}{p}\left(\frac{\varphi(x)^p}{p} - \frac{x^{p^2}}{p^p}\right) = \underbrace{\frac{(x^p + p\delta(x))^p}{p^2}} - \frac{x^{p^2}}{p^{p+1}}$$

$\in A$ because $= p^{p-2}\underbrace{\left(\frac{x^p}{p} + \delta(x)\right)^p}_{\in A}$

Finish by induction using $\gamma_{k \cdot p}(y) = \text{unit}\ \gamma_k(\gamma_p(y))$. $\square$

Lecture 2 § General properties of (derived) prismatic cohomology

Fix prime p

Def. A prism is a pair (A, I) of a δ-ring A, & an invertible ideal $I \subset A$ s.t.

1) A is (derived) (p, I)-complete

2) $p \in (I, \varphi(I))$ ($\Leftrightarrow$ after some localization of A, $I = (d)$, with $\delta(d) \in A^\times$)

Examples 1) A p-complete, p-tors. free, $I = (p)$

$\rightsquigarrow$ (A, I) prism

"crystalline case"

Note: $\delta(p) = \frac{p - p^p}{p} = 1 - p^{p-1} \in A^\times$

2) k perfect field, $A := W(k)[[u]]$, $\delta(u) = 0$. ($\Leftrightarrow$ $\varphi(u) = u^p$)

$I = (E(u))$, $E(u) \in W(k)[u]$ Eisenstein polynomial

"Breuil-Kisin case"

Note: $A/I \simeq \mathcal{O}_K$ for K p-adic field

$u \mapsto \pi$

3) $A = \mathbb{Z}_p[[q-1]]$, $\varphi(q) = q^p$, $I = \left([p]_q = \frac{q^p - 1}{q - 1} = 1 + q + \cdots + q^{p-1}\right)$

"cyclotomic prism"

$A/I \simeq \mathbb{Z}_p[\zeta_p]$, $q \mapsto \zeta_p$

Def. (A, I) prism is

1) bounded if A/I has bounded p^∞-torsion

2) orientable if I principal

3) perfect if $\varphi : A \xrightarrow{\sim} A$

4) transversal if A/I is p-torsion free

3) $(A, I) \to (B, J)$ (faithfully) flat if $A \to B$ is (p, I)-completely (faithfully) flat.

i.e. $A/^{\mathbb{L}}_{(p,I)} \to B/^{\mathbb{L}}_{(p,I)}$ (faithfully) flat.

Remark: - (A, I) perfect iff $A = W(R)$; R perfect $\mathbb{F}_p$-alg

and $I = (\xi)$, $\xi = [a_0] + p[a_1] + \cdots$ and R a_0-adically complete.

$a_1 \in R^\times$.

- $\{$perfect prisms$\} \xleftrightarrow{\sim} \{$perfectoid rings$\}$,

$$(A, I) \longmapsto A/I$$

$$(A_{inf}(T) = W(T^\flat), \ker\theta) \longleftarrow\!\shortmid T$$

<u>Def'n</u> (prismatic site)

1) ("relative") (A, I) bounded prism, R A/I-alg.

$(R/A)_\Delta$ has objects $(R \xrightarrow{\iota} B/J \leftarrow B)$, (B, J) bounded prism over (A, I)

& ι morph. of A/I-alg.

morphisms = morph. of prisms over (A, I) comp. w/ ι.

covers = gen. by faithfully flat maps of prisms

Set $\mathcal{O}_\Delta(R \to B/J \leftarrow B) = B$, $\bar{\mathcal{O}}_\Delta(R \to B/J \leftarrow B) = B/J$

2) ("absolute") Same without base prism (A, I)

Notation: $(R)_\Delta$.

<u>Lemma</u> (rigidity) $(A, I) \to (B, J)$ morph. of prisms. Then $I \cdot B = J$.

<u>Proof</u> Reduce to $I = (d)$, $J = (d')$, $d = u \cdot d'$, $u \in B$.

Then $\underset{\substack{\uparrow \\ B^\times}}{\delta(d)} = u^p \delta(d') + \underbrace{(d')^p \delta(u) + p\delta(u)\delta(d')}_{\in \operatorname{rad} B}$

$\Rightarrow u \in B^\times$. $\square$

<u>Corollary</u>. $(R/A)_\Delta$, $(R)_\Delta$ have non-empty products, calculated by prismatic envelope

<u>Th.</u> (Bhatt-Scholze) R smooth over A/I, (A, I) bounded prism

$$R\Gamma_\Delta(R/A) := R\Gamma((R/A)_\Delta, \mathcal{O}_\Delta) \in \mathrm{CAlg}(\hat{D}(A))$$
$$\circlearrowleft \varphi$$

1) ("cryst. comp") If $I = (p)$, then

$$R\Gamma_{\mathrm{crys}}(R/A) \simeq R\Gamma_\Delta(R/A) \overset{\mathbb{L}}{\underset{A}{\otimes}}{}_\varphi A$$

2) ("Hodge-Tate comp") $H^*((R/A)_\Delta, \widetilde{\mathcal{O}_\Delta}) \simeq \Omega^*_{R/(A,I)}\{-*\}$

$$\swarrow (-) \underset{A/I}{\otimes} (I/I^2)^{-*}$$

$$\left(0 \to I/I^2 \to \mathcal{O}_\Delta/I^2 \to \mathcal{O}_\Delta/I \to 0\right)$$

3) ("de Rham comp") $R\Gamma_{dR}(R/(A,I)) \simeq \left(R\Gamma_\Delta(R/A) \overset{\hat{\mathbb{L}}}{\underset{A}{\otimes}}{}_\varphi A\right) \overset{\hat{\mathbb{L}}}{\underset{A}{\otimes}} A/I$

4) ("étale cohom") Assume A perfect. Let X_η = gen. fibre of $\operatorname{Spf} R$.

$$\Rightarrow R\Gamma_{\text{ét}}(X_\eta, \mathbb{Z}/p^n) \simeq \left(R\Gamma_\Delta(R/A) \overset{\mathbb{L}}{\underset{A}{\otimes}} A/p^n[\tfrac{1}{I}]\right)^{\varphi = 1}$$

5) $R\Gamma_\Delta(R/A)$ commutes with base change in (A, I).

Key technical input: Assume $I = (d)$ and P (p, I)-compl. flat δ-A-alg. $J = (d, x_1, \dots, x_r) \subset P$ with $x_1, \dots, x_r$ (p, d)-completely reg. Then the prismatic envelope $P\{\frac{J}{I}\}^\wedge$ of P in J is $P\{\frac{x_1}{d}, \dots, \frac{x_r}{d}\}^\wedge_{(p,I)}$

Moreover, $P\{\frac{J}{I}\}^\wedge$ is (p,I)- completely flat over A & commutes with base change.

Example (q- de Rham cohom.):

$$A = \mathbb{Z}_p[\![q-1]\!] \quad , \quad R = \mathbb{Z}_p\langle T^{\pm 1}\rangle \quad , \quad \varphi(q) = q^p$$

$$\cup$$

$$I = ([p]_q) \qquad [n]_q = \frac{q^n - 1}{q-1} = 1 + q + \cdots + q^{n-1}$$

Set $R^{(1)} := R \otimes_{\mathbb{Z}_p} A/I \simeq \mathbb{Z}_p[\zeta_p]\langle T^{\pm 1}\rangle$ & $\tilde{R} = A\langle T^{\pm 1}\rangle$, $\varphi(T) = T^p$

<u>Th.</u> (Bhatt- Scholze)

$\Delta_{R^{(1)}/A} \simeq$ q- de Rham complex of $\tilde{R}$ over A

$$= [\tilde{R} \xrightarrow{\nabla_q} \tilde{R}\, d_q T] \quad \text{with} \quad \nabla_q(f(T)) := \frac{f(qT) - f(T)}{qT - T}\, d_q T$$

$$\nabla_q(T^n) = [n]_q T^{n-1} d_q T$$

$$\simeq \left[\hat{\bigoplus_{n\in\mathbb{Z}}} A\cdot T^n \xrightarrow{[n]_q} \hat{\bigoplus_{n\in\mathbb{Z}}} A\cdot T^n \underbrace{\frac{d_q T}{T}}_{=: d_q \log T}\right]$$

<u>Rmk.</u> - de Rham comp. follows $n \equiv [n]_q \mod q-1$

- HT comp. is also explicit.

<u>Note</u>. $\nabla_q(f\cdot g(T)) = f(qT)\,\nabla_q(g(T)) + g(T)\,\nabla_q(f(T))$

Set $A_\infty = \left(\varinjlim_{\varphi} A\right)^\wedge = \left(\bigcup_{n\geq 1} \mathbb{Z}_p[\![q^{1/p^n} - 1]\!]\right)^\wedge_{(p,I)} = A_{inf}(\mathbb{Z}_p^{cycl})$

$$q \longmapsto [\varepsilon], \quad \varepsilon = (1, \zeta_p, \cdots)$$

Note: $(q-1) \in L := A_\infty/p[\frac{1}{I}]$ invertible because $[p]_q \equiv (q-1)^{p-1} \mod p$.

$$\Rightarrow \quad \Delta_{R^{(1)}/A} \overset{L}{\otimes}_A L \simeq \left[\hat{\bigoplus}_{n \in \mathbb{Z}} L \cdot T^n \xrightarrow{q^n - 1} \hat{\bigoplus}_{n \in \mathbb{Z}} L \cdot T^n \frac{dq \log T}{q-1} \right] =: K^\bullet$$

Now, 1) $\varphi(dq\, T) = dq\, T^p = [p]_q\, T^{p-1}\, dq\, T \rightsquigarrow \varphi\left(\frac{dq \log T}{q-1}\right) = \frac{dq \log T}{q-1}$.

2) $$(K^\bullet)^{\varphi=1} \underset{\substack{\uparrow \\ \text{completed colimit} \\ \text{over } \varphi}}{\simeq} \left[\hat{\bigoplus}_{n \in \mathbb{Z}[\frac{1}{p}]} L \cdot T^n \xrightarrow{q^n - 1} \hat{\bigoplus}_{n \in \mathbb{Z}[\frac{1}{p}]} L \cdot T^n \frac{dq \log T}{q-1} \right]^{\varphi=1}$$

$$\underset{\substack{\uparrow \\ \text{almost} \\ \text{purity}}}{\simeq} \left(R\Gamma_{\text{proét}}(X_\eta, \mathcal{O}^b) \right)^{\varphi=1} \underset{\substack{\uparrow \\ \text{Artin-Schreier} \\ \text{sequence}}}{=} R\Gamma_{\text{ét}}(X_\eta, \mathbb{F}_p)$$

Next: Derived prismatic cohomology

<u>Definition</u>. (A, I) bounded prism.

$\Delta_{/A}$: $\{$ derived p-complete animated $A/I \} \longrightarrow \hat{D}(A)$ as

left Kan extension of $R\Gamma_\Delta(-/A)$ on p-compl. smooth A/I-alg.

Then 1) left Kan extending $\tau^{\leq n} R\Gamma_\Delta(-/A)$ yields an increasing exhaustive $\mathbb{N}$-indexed filtration $\mathrm{Fil}_n^{\text{conj}}(\overline{\Delta}_{R/A})$ with $\mathrm{gr}_n^{\text{conj}}(\overline{\Delta}_{R/A}) \simeq \Lambda^n(\hat{L}_{R/(A/I)})\{-n\}[-n]$

2) $R \longmapsto \Delta_{R/A}$ satisfies quasi-syntomic descent

<u>Def'n</u>. R p-complete with bounded p^∞-torsion.

1) R - quasi-syntomic if $\hat{L}_{R/\mathbb{Z}_p}$ of p-compl. Tor-ampl. in $[-1, 0]$

2) $R \to R'$ quasi-syntomic if $\hat{L}_{R'/R}$ — " —

& $R \to R'$ is p-compl. flat.

Lemma $R \to R'$ quasi-syn. cover with Čech nerve $R^\bullet$ & $A/I \to R$ q-syn.

Then $\Delta_{R/A} \simeq \varprojlim_{\Delta} \Delta_{R'/A}$

Powerful strategy to access $\Delta_{-/A}$

1) Reduce to smooth A/I-alg using left Kan-ext

2) Reduce to large A/I-alg R via quasi-syntomic descent

$\Uparrow$

$A/I \to R$ quasi-syntomic & $\exists$ surjection $A/I\langle x_j^{1/p^\infty} : j \in J\rangle \twoheadrightarrow R$

($\rightsquigarrow \hat{L}_{R/(A/I)}[-1]$ p-completely flat module over R)

Consequence: $\Delta_{R/A}$ is concentrated in degree 0 & initial in $(R/A)_\Delta$.

Sample application: Nygaard filtration

Th. R A/I-alg

$\rightsquigarrow \exists$ nat'l filtration Fil_N^i on $\Delta^{(1)}_{R/A} = \Delta_{R/A} \otimes_{A,\varphi} A$

s.t. $\mathrm{gr}_N^i \Delta^{(1)}_{R/A} \simeq \mathrm{Fil}_i^{\mathrm{conj}}(\bar{\Delta}_{R/A})\{i\}$ décalage

If R smooth over A/I, then $\varphi_{R/A} : \Delta^{(1)}_{R/A} \xrightarrow{\sim} L\eta_I \Delta_{R/A}$.

Sketch If R large, set $\mathrm{Fil}_N^i \Delta^{(1)}_{R/A} := \varphi^{-1}(I^i \Delta_{R/A})$

In particular, get de Rham comparison. R smooth over A/I,

$$\Delta_{R/A} \otimes_{A,\varphi} A/I \simeq (L\eta_I \Delta_{R/A}) \otimes_A A/I \simeq \mathcal{H}^*(\bar{\Delta}_{R/A})\{*\} \simeq \Omega^*_{R/(A/I)}$$

with de Rham diff.

Lecture 3 § Prismatization

Fix prime p.

Question. What is the "spectrum of the initial prism"? ($\mathbb{Z}$)

Set $A^\circ = \mathbb{Z}_p\{x, \delta(x)^{-1}\}^\wedge_{(p,x)}$ = initial oriented prism

$\cup$

$I^\circ = (x)$

Expect. flat surjection $\mathrm{Spf}\, A^\circ \to \mathbb{Z}$

Expect. $\mathrm{Spf}\, A^\circ \times_{\mathbb{Z}} \mathrm{Spf}\, A^\circ \simeq \mathrm{Spf}\, A^\circ \times \mathrm{Spf}\left(\mathbb{Z}_p\{u^{\pm 1}\}^\wedge_p\right)$

Lemma. 1) As functors on rings

$$\mathrm{Spec}\, \mathbb{Z}\{u\} = W, \quad \mathrm{Spec}\, \mathbb{Z}\{u^{\pm 1}\} = W^\times$$

2) As functors on $\mathrm{Nilp} := \{R \text{ p-nilpotent}\}$

$$\mathrm{Spf}\, A^\circ_{(R)} = \left\{ \sum_{i=0}^{\infty} V^i[a_i] : a_0 \in R \text{ nilp.}, a_1 \in R^\times \right\}$$

Proof 1) R ring

$$\mathrm{Hom}_{\text{rings}}(\mathbb{Z}\{u\}, R) \simeq \mathrm{Hom}_{\delta\text{-rings}}(\mathbb{Z}\{u\}, W(R)) \simeq \mathrm{Hom}_{\text{rings}}(\mathbb{Z}\{u\}, W(R)) = W(R).$$

rest follows

(A δ-ring, $\varphi(x) = x^p + p\delta(x)$

$$\varphi^n(x) = x^{p^n} + p\,\delta(x)^{p^{n-1}} + p^2 \delta_2(x)^{p^{n-2}} + \cdots + p^n \delta_n(x)$$

then $A \to W(A)$

$a \mapsto (a, \delta(a), \delta_2(a), \ldots))$

Def'n $W\mathrm{Cart} := [\mathrm{Spf}\, A^\circ / W^\times]$

Lemma R adic $/\mathbb{Z}_p \rightsquigarrow \mathrm{WCart}(R) := \mathrm{Hom}(\mathrm{Spf}\, R, \mathrm{WCart})$

$\simeq \{ (I \xrightarrow{\alpha} W(R)) : I \in \mathrm{Pic}(W(R)),$ locally $\alpha = [a_0] + V[a_1] + \cdots \}$

a_0 nilp, $a_1 \in R^\times$

$=$ "Cartier-Witt divisors"

Given (A, I) prism $\rightsquigarrow A \xrightarrow{id} A$ lifts to $\gamma : A \to W(A)$

$a \mapsto (a, \delta(a), \cdots)$

Def'n $\rho_A : \mathrm{Spf}\, A \to \mathrm{WCart}$ given by $(I \otimes_A W(A) \to W(A))$

Note: If $(d) = I$, then $\gamma(d) = (d, \delta(d), \ldots)$

Prop. 1) $(A, I), (B, J)$ bounded prisms, (A, I) transversal,

(C, K) their product in $(\mathbb{Z}_p)_\Delta$,

Then

$$\begin{tikzcd} \mathrm{Spf}\, C \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathrm{Spf}\, B \arrow[d, "\rho_B"] \\ \mathrm{Spf}\, A \arrow[r, "\rho_A"] & \mathrm{WCart} \end{tikzcd}$$

2) (A, I) transversal, $A \neq 0$, then ρ_A is faithfully flat.

3) $D(\mathrm{WCart}) := \varprojlim_{\substack{\mathrm{Spec}\, R \to \mathrm{WCart} \\ R \in \mathrm{Nilp}}} D(R) \simeq \varprojlim_{(A, I)} \hat{D}(A).$

$=$ "prismatic crystals on $(\mathbb{Z}_p)_\Delta$"

In part. $R\Gamma(\mathrm{WCart}, \mathcal{O}) \simeq R\Gamma((\mathbb{Z}_p)_\Delta, \mathcal{O}_\Delta)$

Proof. 1) WLOG $A = B = A^0$.

Then $\mathrm{Spf}\, C \simeq \mathrm{Spf}\, A^0 \times \mathrm{Spf}(\mathbb{Z}_p\{u^{\pm 1}\}_p^\wedge)$

2) Consequence of the key technical statement from yesterday on prismatic envelope

3) follows from flat descent.

In particular. $R\Gamma((\mathbb{Z}_p)_\Delta, \bar{\mathcal{O}}_\Delta) \simeq R\Gamma(\mathrm{WCart}^{HT}, \mathcal{O})$

$$\mathrm{WCart}^{HT} \subset \mathrm{WCart} \quad \text{locus where } a_0 = 0$$

$$\mathrm{WCart}^{HT} \times_{\mathrm{WCart}\, P_A} \mathrm{Spf}\, A \simeq \mathrm{Spf}\, A/I$$

<u>Lemma</u> $\eta : \mathrm{Spf}\, \mathbb{Z}_p \xrightarrow{V(1)} \mathrm{WCart}^{HT}$ faithfully flat.

<u>Proof</u> Take $\alpha = V(u)$, $u \in W^\times(R)$.

Note $V(u) \cdot x = V(F(x) \cdot u)$

$\rightsquigarrow$ Now, $F : W^\times \to W^\times$ faithfully flat. $\square$

<u>Lemma</u> 1) $\underline{\mathrm{Aut}}(\eta) = W^\times[F] = \{x \in W^\times : F(x) = 1\}$

2) $\ker(W \xrightarrow{V(1)} W) = W[F] = \{x \in W : F(x) = 0\}$

<u>Proof</u> $x\, V(1) = V(F(x) \cdot 1)$

<u>Proposition</u> 1) $W^\times[F] \simeq \mathbb{G}_m^\sharp = \mathrm{Spf}\left(\widehat{\bigoplus_{n \geq 0}} \mathbb{Z}_p \cdot \frac{(t-1)^n}{n!}\right)$

$$=: \text{PD-hull of unit section } \mathbb{G}_m :$$

In particular, $\mathrm{WCart}^{HT} \simeq B\mathbb{G}_m^\sharp$.

2) $W(F) \simeq \mathbb{G}_a^\sharp = \mathrm{Spf}\left(\widehat{\bigoplus_{n \geq 0}} \mathbb{Z}_p \cdot \frac{a^n}{n!}\right)$

Proof (only $\geq$) :

$$\begin{tikzcd}
 & W(F) \arrow[dl] \arrow[d] \arrow[r] & \mathrm{Spf}\,\mathbb{Z}_p \arrow[d, "0"] \\
\mathbb{G}_a & \mathbb{G}_a^{\#} \arrow[l] & W \arrow[r, "F"] & W
\end{tikzcd}$$

$$\rightsquigarrow \quad W(F) \simeq \mathrm{Spf}\Big(\mathbb{Z}_p\{y\}^{\wedge} \otimes_{\mathbb{Z}_p\{x\}} \mathbb{Z}_p \Big)$$

$$y(y) \leftarrow\!\shortmid x \qquad x \mapsto 0, \quad x \mapsto \delta(x)$$

$$\simeq \mathrm{Spf}\Big(\mathbb{Z}_p[y_0, y_1, \cdots]^{\wedge}_p \big/ (y_0^p + p y_1,\ y_1^p + p y_2, \cdots) \Big) \supset \mathbb{G}_a^{\#}$$

<u>Prop</u>

1) $D(B\mathbb{G}_m^{\#}) = \{ (M \in \hat{D}(\mathbb{Z}_p),\ \theta : M \to M) : \theta^p - \theta \text{ loc. nilp on } H^*(M \otimes_{\mathbb{Z}_p} \mathbb{F}_p) \}$

2) $D(B\mathbb{G}_a^{\#}) = \{ (M, \theta) : \theta \text{ loc. nilp. on } H^*(M \otimes_{\mathbb{Z}_p} \mathbb{F}_p) \}$

<u>Proof</u> (only $\geq$) for vector bundles:

Set $A = \mathcal{O}_{\mathbb{G}_a^{\#}}$ $\mathbb{Z}_p$- Hopf algebra

$B = \mathrm{Hom}_{\mathbb{Z}_p}(A, \mathbb{Z}_p) \simeq \mathbb{Z}_p[[\theta]]$ is a power series alg.

If M finite free $\mathbb{Z}_p$- module, then equiv.

- a coaction $M \to M \otimes_{\mathbb{Z}_p} A \simeq \mathrm{Hom}_{\mathrm{cont}}(B, M)$
- an action $B \otimes_{\mathbb{Z}_p} M \to M$ s.t. θ is top. nilp. on M. $\square$

<u>Rmk</u> $0 \to \mu_p \to \mathbb{G}_m^{\#} \xrightarrow{\log} \mathbb{G}_a^{\#} \to 0$

<u>Corollary</u> $R\Gamma((\mathbb{Z}_p)_\Delta, \overline{\mathcal{O}}_\Delta) = R\Gamma(W(\mathrm{Cart}^{HT}, \theta) \simeq R\Gamma(B\mathbb{G}_m^{\#}, \mathbb{Z}_p) = (\mathbb{Z}_p \xrightarrow{0} \mathbb{Z}_p)$

R p-complete, $(R \to B/J \leftarrow B) \in (R)_{\Delta}$

Let $\gamma : B \to W(B)$ δ-map lifting Id_B

$\rightsquigarrow$ get $R \to B/J \to B/J \otimes^{\mathbb{L}}_{B,\gamma} W(B) = \mathrm{cone}\left(J \otimes_B W(B) \to W(B)\right)$

morph. of animated rings

<u>Def'n</u> (prismatization) 1) $R^{\Delta} = \mathrm{Wcart}_R :=$ stack on Nilp with

$$R^{\Delta}(S) = \left\{ \left(\underset{\mathrm{Wcart}(S)}{\underset{\uparrow}{I \xrightarrow{\alpha} W(S)}} , \underset{\substack{\uparrow \\ \text{morph. of} \\ \text{animated rings}}}{R \to \mathrm{cone}(\alpha)} \right) \right\}$$

In part, $\mathbb{Z}_p^{\Delta} = \mathrm{Wcart}$.

2) (A, I) prism, R A/I-alg. Then

$$\begin{tikzcd} (R/A)^{\Delta} := \mathrm{Wcart}_{R/A} \arrow[r] \arrow[d] & \mathrm{Wcart}_R = R^{\Delta} \arrow[d] \\ \mathrm{Spf}\, A \arrow[r, "PA"] & \mathrm{Wcart} = \mathbb{Z}_p^{\Delta} \end{tikzcd}$$

<u>Thm</u> (Bhatt-Scholze)

1) R quasi-regular semiperfectoid (R qsyn, $\exists R_0 \twoheadrightarrow R$, R_0 perf'd)

$\rightsquigarrow R^{\Delta} \simeq \mathrm{Spf}(\Delta_R)$ $\quad (\Delta_R = \Delta_{R/A_{\mathrm{inf}}(R_0)})$

2) R qsyn. Then $D(R^{\Delta}) \simeq \{$prismatic crystals on $(R)_{\Delta}\}$. In particular, $R\Gamma((R)^{\Delta}, \mathcal{O}) \simeq R\Gamma((R)_{\Delta}, \mathcal{O}_{\Delta})$.

3) (A, I) bounded prism, $A/I \to R$ quasi-syntomic

$$\rightsquigarrow D((R/A)^{\Delta}) \simeq \{\text{prismatic crystals on } (R/A)_{\Delta}\}$$

<u>Why is prismatization useful?</u>

Have natural ring stack $\mathcal{R}$ over $\mathrm{WCart} = \mathbb{Z}_p^{\Delta}$ with

$$\mathcal{R}(S) := \mathrm{Cone}(\alpha) \quad \text{for } (I \xrightarrow{\alpha} W(S)) \in \mathrm{WCart}(S)$$

In fact, $\mathcal{R} = (\mathbb{Z}_p[x])^{\Delta}$

Given R p-complete, then

$$\underset{\substack{\uparrow \\ \mathrm{Nilp}_{\mathrm{WCart}}}}{R^{\Delta}(S)} = \mathrm{Hom}_{\substack{\text{ring} \\ \text{groupoids}}}(R, \mathcal{R}(S))$$

Set $R^{dR} := \mathrm{Spf}\,\mathbb{Z}_p \times_{\mathrm{WCart}} \mathcal{R}$ with $\mathrm{Spf}\,\mathbb{Z}_p \xrightarrow{p} \mathrm{WCart}$ $\quad \in W(\mathbb{Z}_p) \to W(\mathbb{F}_p)$, $p = V(1)$

<u>Lemma</u>. $\mathcal{R}^{dR} \simeq [\mathbb{G}_a / \mathbb{G}_a^{\#}]$ over $\mathrm{Spf}\,\mathbb{Z}_p$

<u>Proof</u>. $\mathbb{G}_a / \mathbb{G}_a^{\#} \xleftarrow{\sim} W/VW \oplus \mathbb{G}_a^{\#} \xrightarrow{\sim} W/FVW \simeq W/p = \mathcal{R}^{dR}$

$\mathbb{G}_a \simeq W/VW$

$\underset{\substack{\uparrow \\ F: W \to W \\ \text{faithfully flat} \\ \mathbb{G}_a^{\#} = W[F]}}{}$

Corollary: k perf. field, $k \to R$ smooth

Then $\varphi^*_{W(k)} R\Gamma_{\Delta}(R/W(k)) \simeq R\Gamma_{\mathrm{crys}}(R/W(k))$

Proof. $R\Gamma_{\Delta}(R/W(k)) \simeq R\Gamma((R/W(k))^{\Delta}, \mathcal{O})$

$$\simeq R\Gamma(\underbrace{\mathrm{Hom}(R, \mathcal{R}^{dR}_{W(k)})}_{\text{stack on } \mathrm{Nilp}_{W(k)}}, \mathcal{O})$$

$$\simeq R\Gamma(\mathrm{Hom}(R, [\mathbb{G}_a/\mathbb{G}_a^{\#}]), \mathcal{O})$$

If $\tilde{R}$ is a smooth lift of R to $W(k)$, then

$$\mathrm{Spf}\,\tilde{R} \underset{\text{flat surj}}{\twoheadrightarrow} \mathrm{Hom}(R, \mathbb{G}_a/\mathbb{G}_a^{\#})$$

with Čech nerve given by PD-envelopes of the diagonals

Thus. $\simeq R\Gamma_{crys}(R/W(k))$

Another application: (A, I) bounded prism

Set $\mathcal{R}^{HT} := \mathrm{Spf}\,A/I \times_{W\mathrm{Cart}} \mathcal{R} \simeq \mathrm{Spf}\,A/I \times_{W\mathrm{Cart}^{HT}} \mathcal{R}_{/W\mathrm{Cart}^{HT}}$

<u>Lemma</u> $\exists$ natural morphism of ring stacks

$\mathcal{R}^{HT} \longrightarrow \mathbb{G}_a$ whose kernel is $\mathbb{G}_a^{\#}\{1\}[1]$. which is square zero.

<u>Proof</u> $\pi_0(\mathcal{R}^{HT}(S)) = W(S)/V(1)W(S) \to S = W(S)/VW(S)$

$V(1)\cdot W(S) = V(F(W(S)))$

$$\begin{tikzcd} M \arrow[r] & C \arrow[r] & C_0 \\ & C \arrow[r] \arrow[d] & C_0 \arrow[d, "(id,0)"] \\ & C_0 \arrow[r, "(id,\delta)"'] & C_0 \oplus M[1] \end{tikzcd}$$

$\pi_1(\mathcal{R}^{HT}(S)) \simeq I/I^2 \otimes_{A/I} W[F](S) \simeq \mathbb{G}_a^{\#}\{1\}(S)$

<u>Cor.</u> Assume $R \to A/I$ smooth. Set $\bar{A} = A/I$. Then $(R/A)^{HT} \longrightarrow \mathrm{Spf}\,R$. is a gerbe banded by $\tau^{\#}_{R/\bar{A}}\{1\}$ = PD hull of zero section in $\tau_{R/\bar{A}}\{1\} = \mathrm{Spf}(\mathrm{Sym}_R(\hat{\Omega}^1_{R/\bar{A}}[-1]))$

Proof. derived deformation theory

Corollary. $H^i((R/A)_\Delta, \mathcal{O}_\Delta) \simeq \hat{\Omega}^i_{R/\bar{A}}\{-i\}$

$$0 \xrightarrow{0} \Omega^1_{R/\bar{A}}\{-1\} \xrightarrow{0} \cdots$$

$$D(B\tau^{\mathbb{N}}_{R/\bar{A}}\{1\}) \simeq \{(M + \hat{D}(R),\ \theta : M \to \hat{\Omega}^1_{R/\bar{A}}\{-1\} \otimes_R M)\ \ \theta \wedge \theta = 0\}$$

θ top. nilp.

$$\subset D(\mathrm{Sym}^\cdot(T_{R/\bar{A}}\{1\}))$$

Lecture 4. The prismatic logarithm

(A, I) any prism.

Aims.

1) Construct a natural invertible A-module $A\{1\}$ with the natural data:

- an isom $\varphi_{A\{1\}} : \varphi^* A\{1\} \xrightarrow{\sim} I^{-1}A\{1\}$ (write $\varphi_{A\{1\}} : A\{1\} \xrightarrow{\text{can}} \varphi^* A\{1\} \xrightarrow{\varphi_{A\{1\}}} I^{-1}A\{1\}$)
- an isom $A\{1\} \otimes_A A/I \simeq I/I^2$
- if $I = (p)$, a trivialization $A \simeq A\{1\}$.

Heuristic: $A\{1\}$ "=" $I \otimes_A \varphi^* I \otimes_A \cdots$

2) Construct a natural homomorphism

$$\log_\Delta : (1+I)^{rk=1} \longrightarrow A\{1\}$$

$$\| \qquad\qquad\qquad \text{s.t.}$$

$$\{x \in 1+I : \delta(x) = 0\}$$

- $\varphi_{A\{1\}}(\log_\Delta(x)) = \log_\Delta(x)$
- $\log_\Delta(x) \equiv x - 1 + I/I^2$

Consequences : - Breuil-Kisin twists

$$M\{j\} = M \otimes_A A\{j\} \quad , j \in \mathbb{Z}$$

- Set $\mathcal{N}^{\geq \cdot} A := \varphi^{-1}(I^{\cdot})$

$$\rightsquigarrow \varphi_j : \mathcal{N}^{\geq j} A\{j\} \longrightarrow A\{j\}, \quad a \otimes b \mapsto \varphi(a) \cdot \varphi_{A\{j\}}(b)$$

Concretely, assume $I = (d)$, $A\{1\} = A \cdot e$, $\varphi_{A\{1\}}(e) = \frac{1}{d} \cdot e$

$$\rightsquigarrow \mathcal{N}^{\geq j} A \overset{e^j}{\simeq} \mathcal{N}^{\geq j} A\{j\} \xrightarrow{\varphi_j} A\{j\} \overset{e^j}{\simeq} A$$

$$a \longmapsto \frac{\varphi(a)}{d^j}, \quad j \in \mathbb{Z}$$

<u>Note</u>. $\log_\Delta(-) \in \mathcal{N}^{\geq 1} A\{1\}$

Set $R := A / \mathcal{N}^{\geq 1} A \rightsquigarrow$ get $R \longrightarrow A/I$

$$\bar{a} \longmapsto \overline{\varphi(a)}$$

Set $[-] : R^b \longrightarrow (A/I)^b \longrightarrow A$ Teichmüller lift

$$R^b = \varprojlim_{x \mapsto x^p} R$$

$$\rightsquigarrow [T_p R^\times] \subset (1+I)^{rk=1}$$

$$[T_p R^\times] = \{(1, r_1, \cdots) \in R^b\}$$

Get $T_p R^\times \xrightarrow{\log_\Delta([-])} eq\left(\mathcal{N}^{\geq 1} A\{1\} \underset{can}{\overset{\varphi_1}{\rightrightarrows}} A\{1\}\right)$

$$\downarrow$$

$$A\{1\} \otimes_A A/I \simeq I/I^2$$

$$\zeta \longmapsto [\zeta] - 1$$

If (A, I) transversal, then (later) injective.

<u>Example</u>. Assume $(\mathbb{Z}_p[\![q-1]\!], ([p]_q)) \to (A, I)$

$$\rightsquigarrow q^p \in (1+I)^{rk=1}$$

Lemma: $A\{1\} = A \cdot e_A$ with $(q-1)\cdot e_A = \log_\Delta(q^p)$

(later)

$$\varphi_{A\{1\}}(e_A) = [p]_q^{-1} \cdot e_A$$

<u>Definition</u> $$\log_q(u) = \sum_{n=1}^{\infty} (-1)^{n-1} q^{-n(n-1)/2} \frac{(u-1)(u-q)\cdots(u-q^{n-1})}{[n]_q}$$

$$\log_q(u) \equiv \log(u) \mod q-1$$

$$\log_q(u) \text{ "=" } \frac{q-1}{\log q} \log(u)$$

Unique series s.t. $\log_q(1) = 0$ & $\nabla_q(\log_q(u)) = \frac{1}{u}$

i.e. $$\frac{\log_q(qu) - \log_q(u)}{q-1} = 1$$

<u>Lemma</u> $\log_\Delta(u^p) = \log_q(u) \cdot e_A$ for $u \in A^{rk=1}$ s.t. $\varphi(u) \in 1+I$.

<u>Proof</u>. Write $\log_\Delta(u^p) = f(u) \cdot e_A$.

Check $f(1) = 0$.

$$\log_\Delta(q^p \cdot u^p) - \log_\Delta(u^p) = (q-1)\cdot e_A$$

Note, let $\nu(-) : R^\flat \to A$ Teichmüller lift for the surj. $A \twoheadrightarrow R$.

$$\rightsquigarrow \varphi(\nu(-)) = [-].$$

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d] & A \arrow[d] \\ R \arrow[r] & A/I \end{tikzcd}$$

If $x \in T_p R^\times$, $\log_{\mathbb{A}}([x]) = \log_{\mathbb{A}}(\varphi(\nu(x)))$

$$= \log_{\mathbb{A}}(\nu(x)^p) = \log_q(\nu(x))$$

Thus, if R qrsp $\mathbb{F}_p$-alg, $A = \mathbb{A}_R$.

$$\rightsquigarrow \log_{\mathbb{A}}([x]) = \log_q(\nu(x)) \in \mathbb{A}_R^{\varphi = [p]_q} \overset{\frac{\log(q)}{q-1}}{\simeq} \mathbb{A}_R^{\varphi = p}$$

$$\log(\nu(x))$$

If $R = \mathcal{O}_C/p \Rightarrow A_{inf}(\mathcal{O}_C) \to \mathbb{A}_R$

$$q \overset{\in \mathbb{Z}_p[\![q-1]\!]}{\longmapsto} [\varepsilon]$$

Σ

Construction of $A\{1\}$

I) following Drinfeld

Recall $\{$line bundles for $\mathcal{O}_{\mathbb{A}}$ on $(\mathbb{Z}_p)_{\mathbb{A}}\} \simeq \{$line bundle on $\mathrm{WCart}\}$

Note: Have $F: \mathrm{WCart} \to \mathrm{WCart}$ induces $F: W \to W$

Want: $\mathcal{O}\{1\} \in \mathrm{Pic}(\mathrm{WCart})$ + Isom $F^*\mathcal{O}\{1\} \simeq \mathcal{I}^{-1}\mathcal{O}\{1\}$ with $\mathcal{I}$ = ideal sheaf of WCart^{HT}.

<u>Lemma</u>. $\mathbb{Z}_p \simeq H^0(\mathrm{WCart}, \mathcal{O})$

$$(\mathbb{Z}_p[\![q-1]\!]^{\mathbb{Z}_p^\times} \simeq \mathbb{Z}_p)$$

Set $p := p_{\mathbb{Z}_p} : \mathrm{Spf}\,\mathbb{Z}_p \xrightarrow{p} \mathrm{WCart}$

<u>Corollary</u>. $\mathrm{Pic}^\natural = \{\mathcal{L}$ line bundle on WCart + triv. $\mathbb{Z}_p \simeq p^*\mathcal{L}\}$ is equiv. to a set.

Example: $\mathcal{I}$ naturally defines an object using $p^*\mathcal{I} \simeq (p) \xhookrightarrow[\sim]{p} \mathbb{Z}_p$

Lemma $\mathrm{Id} - F^* : \mathrm{Pic}^1 \xrightarrow{\sim} \mathrm{Pic}^1, \quad \mathcal{L} \mapsto \mathcal{L} \otimes F^* \mathcal{L}^{-1}$.

Thus, $\mathcal{O}(1) := (\mathrm{Id} - F^*)^{-1}(\mathcal{L})$

Proof Given $\mathcal{L}$, set $\bigotimes_{i=0}^{\infty} (F^i)^* \mathcal{L}$

Well-defined: given $S \in \mathrm{Nilp}$, $\mathrm{Spec}\, S \to \mathrm{Wlart}$.

$\rightsquigarrow \exists\ i_0 \geq 0$ s.t. for $i \geq i_0$,

$$\begin{tikzcd} \mathrm{Spec}\, S \arrow[r] \arrow[d, dashed] & \mathrm{Wlart} \arrow[d, "F^i"] \\ \mathrm{Spf}\, \mathbb{Z}_p \arrow[r, "\rho"] & \mathrm{Wlart} \end{tikzcd}$$

factors canonically over

Use 1) Given $\alpha = [a_0] + Vb$ with $a_0^{p^i} = e$

$$\rightsquigarrow F^i(\alpha) = p \cdot \underbrace{F^{i-1} b}_{\in W^\times(S)}$$

2) Exists $i \geq 0$ s.t. $\forall\, u \in W^\times(S)$ with $pu = p$, then $F^i(u) = 1$.

Lemma. $\mathcal{O}(1)|_{\mathrm{Wlart}^{HT}} \simeq \mathcal{I}|_{\mathrm{Wlart}^{HT}}$.

Proof STS:

$$\begin{tikzcd} \mathrm{Wlart}^{HT} \arrow[r] \arrow[d, "\exists\ \mathrm{can}"'] & \mathrm{Wlart} \arrow[d, "F"] \\ \mathrm{Spf}\, \mathbb{Z}_p \arrow[r, "\rho"] & \mathrm{Wlart} \end{tikzcd}$$

Indeed, $FV(u) = p \cdot u$ and $\underline{\mathrm{Aut}}\left(W(S) \xrightarrow{V(u)} W(S)\right) = W^\times[F]$.

II) following Bhatt-Lurie

WLOG (A, I) transversal.

Set $I_r = I \cdot \varphi^* I \cdots (\varphi^*)^{r-1} I$, let $\pi_r : I_r / I_r^2 \to I_{r-1} / I_{r-1}^2$.

Lemma 1) $I_r \subset A$ invertible

2) $A \xrightarrow{\sim} \varprojlim_r A/I_r$, $A/I_r \hookrightarrow \prod_{0\le s<r} A/(\varphi^s)^* I$ $\left(\rightsquigarrow \begin{array}{l} \text{consequence} \\ A\{1\} \ \varphi_{A\{1\}} = 1 \\ \hookrightarrow I/I^2 \end{array} \right)$

3) π_r is div. by p & $\frac{1}{p}\cdot \pi_r \cdot I_r/I_r^2 \to I_{r-1}/I_{r-1}^2$ is surj.

Set $A\{1\} = \varprojlim_r \left(\cdots \to I_r/I_r^2 \xrightarrow{\frac{1}{p}\pi_r} I_{r-1}/I_{r-1}^2 \to \cdots \right)$ invertible A-module

4) $A\{1\} \otimes_A A/I \simeq I/I^2$ via first proj.

5) $A\{1\} \otimes_A A\{\frac{d}{p}\}^\wedge_p$ is can. trivial. $I=(d)$

Proof (sketch) 1) WLOG (p,d) regular

$\rightsquigarrow (p, \varphi^r(d))$ reg

$\rightsquigarrow (\varphi^r(d), p)$ reg.

2) Check $A \simeq \varprojlim A/I_r$ mod p

For rest, $r=2$ (for simplicity) , $I=(d)$

Take $y = \varphi(d)x \Rightarrow \varphi(d)\cdot x \equiv p\cdot \delta(d)x \mod I$

If $y \in I \Rightarrow d \mid x \Rightarrow A/I_2 \hookrightarrow A/I \times A/\varphi^* I$

Now, $I_2 \longrightarrow I/I_2$

$d\cdot\varphi(d) \longmapsto d\left(d^p + p\delta(d)\right) \equiv p\cdot\delta(d)$

4) ✓

5) $I_r \cdot A\{\frac{d}{p}\}^\wedge_p = (p^r) \xrightarrow{\frac{1}{p}} (p^{r-1}) = I_{r-1}\ A\{\frac{d}{p}\}^\wedge_p / I_{r-1}$

Set $\varphi_{A\{1\}} : \varphi^* A\{1\} \xrightarrow{\sim} I^{-1} A\{1\}$ is induced by $\varphi^* I_{r-1} \simeq I^{-1} I_r$.

More precisely, $\varphi^* I_{r-1} \otimes_A A/\varphi^* I_{r-1} \simeq I^{-1} I_r \otimes_A A/\varphi^* I_{r-1}$

$$\varphi^*(I_{r-1}/I_{r-1}^2) \simeq$$

$$\wr\wr$$

$$I^{-1} I_r \otimes_A A/I_r \otimes_{A/I_r} A/\varphi^* I_{r-1}$$

$$\wr\wr$$

$$I^{-1} A\{1\} \otimes_A A/\varphi^* I_{r-1}$$

$\rightsquigarrow$ pass to inv. limit over r.

Next prismatic logarithm

WLOG (A, I) transversal

<u>Proposition</u> $u \in (1+I)^{rh=1}$

1) $u^{p^r} - 1 \in I_{r+1}$

2) $\frac{1}{p} \pi_r \cdot I_{r+1}/I_{r+1}^2 \longrightarrow I_r/I_r^2$ maps $u^{p^r}-1$ to $u^{p^{r-1}}-1$.

$\rightsquigarrow$ Set $\log_{\Delta}(u) := (\underset{\substack{\uparrow \\ I/I^2}}{u-1}, \underset{\substack{\uparrow \\ I_2/I_2^2}}{u^p-1}, \cdots) \in A\{1\}$

3) $\log_{\Delta}(-)$ is a group hom.

4) $\varphi_{A\{1\}}(\log_{\Delta}(-)) = \log_{\Delta}(-)$.

<u>Proof</u> 1) Use $A/I_{r+1} \hookrightarrow \prod_{0 \le s \le r} A/(\varphi^s)^* I$ & $\varphi^s(\underset{\substack{\uparrow \\ I}}{u^{p^{r-s}}-1}) \in (\varphi^s)^* I$

$$u^{p^r}-1$$

2) $x = u^{p^{r-1}} \overset{1)}{\rightsquigarrow} x \equiv 1 \mod I_r$

$\rightsquigarrow (x-1)(\underset{\substack{\| \\ x^p-1 = u^{p^r}-1}}{1+x+\cdots+x^{p-1}}) \equiv \underset{\substack{\| \\ p(u^{p^{r-1}}-1)}}{p(x-1)} \mod I_r^2$

4) $\varphi(u^{p^r}-1) = u^{p^{r+1}}-1$

Assume $\left(\mathbb{Z}_p[\![q-1]\!] . ([p]_q) \right) \longrightarrow (A, I)$

<u>Lemma</u>. $A\{1\} = A \cdot e_A$ with $(q-1)\cdot e_A = \log_{\mathbb{G}}(q^p)$

$\varphi_{A\{1\}}(e_A) = [p]_q^{-1} e_A$

<u>Proof</u> $I_r = \left([p^r]_q = \dfrac{q^{p^r}-1}{q-1} \right)$, define e_A via $\dfrac{q^{p^r}-1}{q-1} \in I_r / I_r^2$. $\square$